

Jeneel Panchal

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SUMMARY

AI/ML Engineer (M.S. Artificial Intelligence, May 2026) specializing in LLMs, Retrieval-Augmented Generation, and end-to-end MLOps. Strong track record of taking models from research to production: fine-tuning with QLoRA/PEFT, engineering retrieval pipelines that minimize hallucination, and building monitored, self-healing deployment systems. Combines deep-learning fundamentals with full-stack and cloud engineering to ship reliable, measurable AI products. Authorized to work in the U.S. for 3 years (OPT + STEM OPT). Proficient in PyTorch, Hugging Face Transformers, LangChain, FastAPI, Docker, AWS, and Azure.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash

ML / DL: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, ONNX, CUDA, Distributed Training

LLMs / GenAI: GPT-4o, Llama 3, BERT, RAG, LangGraph, LangChain, Prompt Engineering, Embeddings, ChromaDB, RAGAS, QLoRA, PEFT, LoRA Fine-Tuning, bitsandbytes, SFT

Computer Vision: OpenCV, CNNs, Vision Transformers (EEViT), Diffusion Models (DDPM), Contrastive Learning

MLOps / Cloud: AWS (EC2, S3, Lambda), Azure (VM, Blueprints, Policy), Docker, GitHub Actions, FastAPI, Redis, Kafka, Weights & Biases, vLLM

Full-Stack: React, Next.js, Node.js, .NET Core, Flask, FastAPI, REST APIs, HTML5, CSS3, Tailwind CSS, Bootstrap, Streamlit

Data: PostgreSQL, MySQL, Supabase, Redis, ChromaDB, Pandas, NumPy

Foundations: Data Structures, Algorithms, System Design, Distributed Systems, Agile/Scrum, SDLC

EXPERIENCE

Adri IT Solutions

Software Engineering Intern (Web Development)

Vadodara, India

Jan 2024 – Jun 2024

- Shipped 8+ responsive production interfaces serving 5,000+ monthly users by implementing component-based HTML/CSS/JS architecture, reducing client-reported UI defects by 40%.
- Cut average page load time by 30% (4.2s → 2.9s) by refactoring 3,000+ lines of legacy CSS/JS, lazy-loading assets, and minifying bundles across 3 client codebases.
- Resolved 15+ cross-browser defects within a 24-hour SLA, maintaining 100% production uptime over a 6-month engagement.

AI/ML PROJECTS

SecureLLM: Fine-Tuned LLM with Dual-Source RAG | Llama 3.1-8B, QLoRA, FastAPI, NVD API

- Fine-tuned Llama 3.1-8B on 9,992 NIST NVD CVE records using QLoRA (rank-64, 4-bit NF4 quantization) over 500 steps on a Colab T4 GPU, producing a domain-adapted model deployable on a single 8GB-VRAM consumer GPU.
- Achieved 100% severity accuracy and 100% CVSS accuracy (± 0.5) across a 7-CVE functional benchmark (Log4Shell, PAN-OS, Outlook NTLM, etc.) by architecting a dual-source RAG pipeline injecting live NVD API facts and a 15-actor MITRE ATT&CK threat-intel KB into the system prompt.
- Reduced time-to-first-token from 80–100s to <3s by implementing Server-Sent Events (SSE) streaming on a FastAPI `/query/stream` endpoint, with a backwards-compatible sync endpoint for batch use.

DocGuard AI: RAG Assistant with Hallucination Defense | Python, LangChain, ChromaDB, Llama 3, RAGAS

- Built as graduate major project under the guidance of Dr. Ausif Mahmood; reduced LLM hallucination rate by 85% on a 50-document PDF benchmark by architecting a 3-layer defense pipeline (retrieval grounding, citation verification, confidence scoring) over a ChromaDB vector store with 384-dim sentence embeddings.
- Cut document QA latency to <1.8s per query by implementing semantic chunking (512-token / 50-token overlap) and local Llama 3.1 8B inference via Ollama, eliminating external API costs.
- Improved retrieval precision@5 by 32% over a vanilla RAG baseline via MMR re-ranking and metadata filtering, validated with RAGAS faithfulness and answer-relevancy metrics.

ML Model Monitor: Production MLOps Pipeline | XGBoost, FastAPI, Evidently AI, PostgreSQL, Docker

- Trained an XGBoost fraud classifier on 284,807 credit-card transactions (0.17% fraud rate) using SMOTE oversampling and StandardScaler, achieving 0.9993 AUC-ROC, 1.00 precision, 0.98 recall, and 0.99 F1 on a time-ordered held-out eval set.
- Built a three-signal drift-monitoring pipeline (data drift on 29 features via Kolmogorov–Smirnov + Evidently AI, concept drift on output distribution, AUC degradation vs. fixed benchmark) running every 60 minutes via APScheduler, with auto-retrain, model hot-swap, and 2-hour cooldown to prevent thrashing.
- Deployed end-to-end on Render via Docker with managed PostgreSQL for prediction logs, monitoring reports, and retrain history; serves predictions at ~120ms latency through a FastAPI REST API and a 7-page Chart.js live dashboard.

EDUCATION

University of Bridgeport

M.S. in Artificial Intelligence GPA: 3.7/4.0

Bridgeport, CT

May 2026

- Coursework:** Deep Learning, Computer Vision, NLP, Cloud Computing, DSA, Software Architecture.

Navrachana University

B.Tech in Computer Science and Engineering

Vadodara, India

Jun 2024